

Modelling Ca^{2+} flow and energetic cost

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1 Introduction

This document describes the modelling of Ca^{2+} flow within the muscle fibre. Here we retain a phenomenological description of the Ca^{2+} concentrations throughout the muscle fibre by following the formulation from [MCL23] and the corresponding parameters within. However, the framework within this document can also be applied to a more detailed Ca^{2+} flow formulation (e.g. 9-state Ca^{2+} model [BCM83; BL22]) provided there exist a rate of flow of Ca^{2+} into the sarcoplasmic reticulum and a description of Ca^{2+} bound to troponin.

The model here assumes a stimulation phase where there is calcium released from the SR. When the muscle is not being stimulated we assume that the Ca^{2+} will be pumped back into the SR. We then model energetic rates based on this rate of pumping into the SR. This is an energy consuming process as the pumping is against the electrochemical potential

$$\overline{\Delta G} = \underbrace{RT \log \left(\frac{[\text{Ca}^{2+}]_{\text{cyto}}}{[\text{Ca}^{2+}]_{\text{SR}}} \right)}_{\text{Chemical potential}} + \underbrace{FzV}_{\text{Electrical potential}}. \quad (1)$$

Here we define $\overline{\Delta G}$ as the free energy over the membrane, T is temperature, $[\text{Ca}^{2+}]_{\text{cyto}}$ and $[\text{Ca}^{2+}]_{\text{SR}}$ are the Ca^{2+} concentrations in the cytoplasm and SR respectively, F is the Faraday constant, V is the resting membrane potential, and z is the charge of the Ca^{2+} ions. While this relationship holds, it may not account for the full description of energetic cost across the membrane; for example the role of other ions in the potential or other processes that relating to Ca^{2+} transport. Thus, we can take a phenomenological approach by using measures of heat released during contractions and using these data to inform the model. Modelling using the above equation would also require a detailed model of the Ca^{2+} dynamics, which would mean an increased level of detail that we can not validate *in vivo*.

The model described in this document is limited by assuming an instant diffusion of Ca^{2+} across the cytoplasm. A diffusion model could be implemented but may not provide a meaningful information for the calculation of energetic cost.

This model is implemented for a single motor unit firing under different firing conditions. Further, at this point the role of shortening and lengthening (and the rate of shortening and lengthening) is not implemented. It is likely that this should be implemented in the same way as the existing models.

2 Methods

The mathematical model of muscle energy use within this study can be split into a Ca^{2+} flow and energetic model. A description Ca^{2+} flow is shown in Figure 1, but within this work, we assume only three states for Ca^{2+} : i) Ca^{2+} in SR, ii) free Ca^{2+} in the cytoplasm, and iii) bound Ca^{2+} to troponin. The implementation is described in the following subsections.

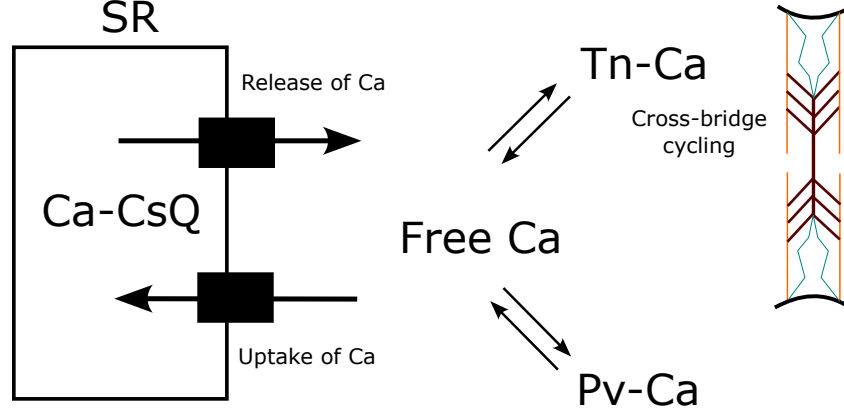


Figure 1: Ca flow in a muscle fibres.

2.1 Ca^{2+} transport

To handle the concentration of Ca ions throughout the membrane we can quantify the Ca transport using a model similar to that described in [MCL23]. The flow of Ca^{2+} can be implemented through an ODE

$$\dot{c}_{ca} = \begin{cases} \frac{1-c_{ca}}{\tau_{act}}, & \text{if } t \in \mathcal{T} \\ -\frac{c_{ca}}{\tau_{deact}}, & \text{otherwise,} \end{cases} \quad (2)$$

where $(\dot{\cdot})$ denotes a time derivative, c_{ca} is the Ca^{2+} concentration within the cytoplasm, τ_{act} and τ_{deact} are the activation and deactivation time constants respectively. $\mathcal{T}$ denotes a time period where the muscle is being stimulated. The concentration of bound Ca^{2+} to Troponin is then calculated as

$$c_{CaTn} = \frac{c_{ca}^n}{c_{ca}^n - c_{ca,50}^n}, \quad (3)$$

where n and $c_{ca,50}$ are constants.

2.2 Energetic cost

To model the energetic cost of the Ca dynamics, we break the energy required to reproduce Ca by accounting for the activation (Ca^{2+} flow) and maintenance (cross-bridge cycling) components separately. Experimental measures have been collected from the mouse soleus and extensor digitorum longus muscle which captures slow and fast fibre-type properties respectively [Bar96]. These experiments have determined isometric heat rate constants r_{am} which accounts for both maintenance and activation rates. Given existing data [BWC07] on the relative heat rates from the two process we assume the rates are split so that 40% is activation

and 60% is maintenance. This accounts for the approximate Q_{10} values from [BWC07], however, no data was taken at 30°C, so it is not clear if we should increase the percentage of activation heat at the higher temperatures.

$$r_a = 0.4 r_{am} \quad (4)$$

$$r_m = 0.6 r_{am}. \quad (5)$$

r_{am} is the activation and maintenance heat rate from [Bar96].

We then compute the maintenance heat rate as

$$\dot{q}_m = r_m c_{CaTn}, \quad (6)$$

where c_{CaTn} is a normalised value of concentration of the CaTn complex assumed to be proportional to cross-bridge cycling.

We compute the activation component by assuming that the $\dot{q}_a$ scales with the rate of Ca pumping back into the SR (we assume negligible contribution from this heat rate during the stimulation period since the duration of stimulus is small). We then express this quantity as

$$\dot{q}_a = \begin{cases} -r_a \dot{c}_{Ca}, & \text{if } \dot{c}_{Ca} < 0, \\ 0, & \text{otherwise.} \end{cases} \quad (7)$$

Within this document the total energetic rate is then assumed to be a combination of the above rates (neglecting recovery for now)

$$\dot{E} = \dot{q}_m + \dot{q}_a. \quad (8)$$

3 Numerical experiments

To test this implementation, a single motor unit from a single subject was tracked across contraction rates of 0.125, 0.25, 0.5, and 1.0 MVC s^{-1} (Figure 2). These measured stimulation times were then used as input to the Ca^{2+} model. The results are shown in Figure 3. The energetic rate increases at the motor unit level across the conditions. This may be a result of the increased time the MU was active during the contractile condition as the firing rates appeared to be fairly similar for this MU (Figure 2).

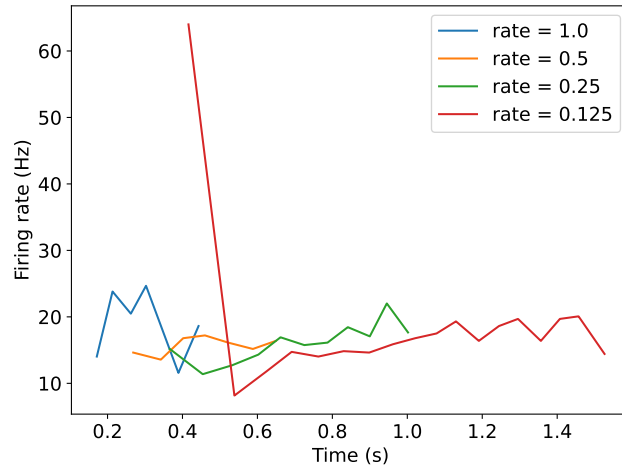
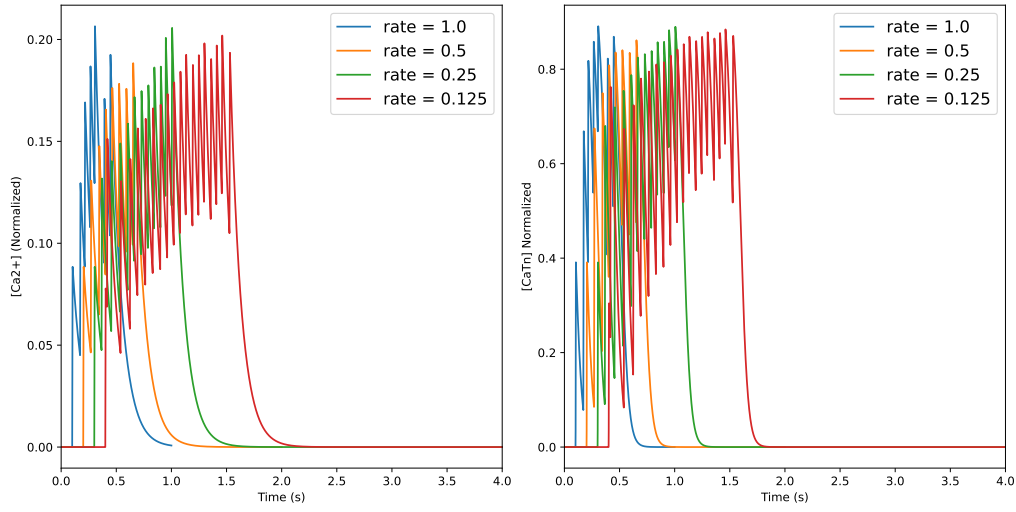
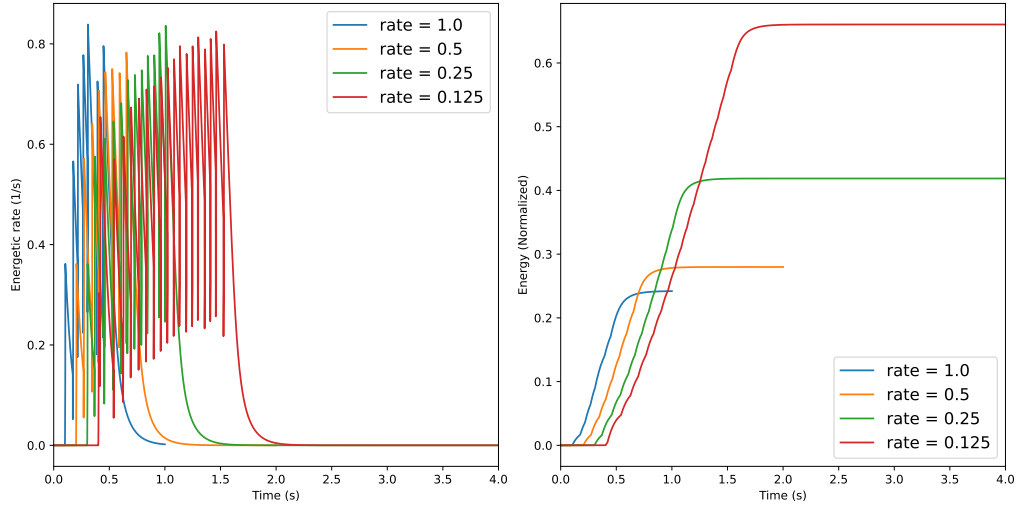


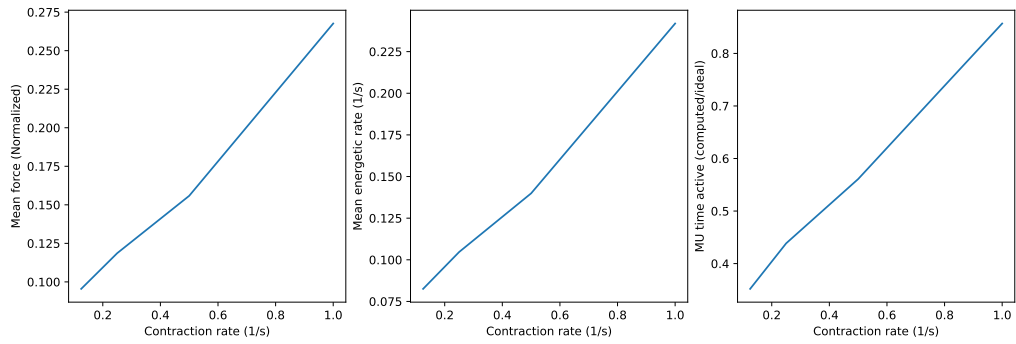
Figure 2: Experimental firing rates of the motor units for the different contraction rates.



(a) *Ca dynamics. Concentration of Ca^{2+} (left) and concentration of bound Ca^{2+} Tn (right)*



(b) *Time-varying energetic rates (left) and energy consumed (right).*



(c) *Mean force produced by the muscle fibre (left). Mean energetic rate computed over a cycle (middle). Ratio of the time between the first and last stimulus to the ideal length of time active (right)(ie. the time to ramp of to full activation, so for the contraction rates of 0.125, 0.25, 0.5, 1.0, the time to full activation (40% MVC) would be 0.4, 0.8, 1.6, and 3.2 s.*

Figure 3: Time-varying Ca concentrations and energetic rates for a single MU at different contraction rates.

4 Appendix: Implementation of energetics

4.1 Implementation of energetics

```
class EnergeticsModel():
    def __init__(self):
        return None

    def actEnergetics(self, t, ca_vec, catn_vec, params):
        """
        Activation component of the energetics
        designed for investigating Ca dependent properties of activation model

        Input
        t: Time (s)
        ca_vec: Ca in the cytoplasm
        catn_vec: Concentration of bound CaTn complex (assume proportional to force)
        params: Parameter dictionary
        """

        # Define constants associated with the activation and mainenance processes
        r_a = 0.4 * params['r_am'] # r_am is the parameter from Barclay 1996
        r_m = 0.6 * params['r_am']

        # Compute rate of ca transport for later calculation
        dt = np.diff(t, prepend=t[0]) # Compute time differences, prepend the first value
                                         to match dimensions
        dt[dt == 0] = 1e-10 # Replace zeros in time differences with a small
                                         epsilon
        dcadt_vec = np.diff(ca_vec, prepend=ca_vec[0]) / dt # Safely compute derivative

        # Compute heat of Ca transport
        q_a = - r_a * dcadt_vec * (dcadt_vec < 0) # 1/s, Use measured quantity scaled based
                                                         on Ca released

        # Compute cross-bridge energetics (maintenance heat rate)
        # This will be scaled based on catn_vec
        q_m = r_m * catn_vec # 1/s, heat rate of cross-bridge kinetics

        return q_a, q_m
```

4.2 Calculation of mean energetic rate, mean force, and time active

```
# Import the model
from Models.MUEnergeticsModel import EnergeticsModel

# Initialize energetics model
energy_model = EnergeticsModel()

# Compute the energetics based on Ca and CaTn
q_a, q_m = energy_model.actEnergetics(t_vec, ca_vec, catn_vec, params['Energetics_model',
                                                                    ])
```

```

117     # Mean energetic rate over time course (q_a,q_m are computed for a full cycle)
118     mean_energy_rates[idx] = np.mean(q_a + q_m)
119
120     # Compute mean force
121     mean_force[idx] = np.mean(catn_vec)
122
123     # Calculate time active
124     time_active[idx] = ((t_stim_idxs[-1] - t_stim_idxs[0]) / fsamp) # s
125

```

References

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